

NEW ULTRA-WIDE-FIELD ANGIOGRAPHIC GRADING SCHEME FOR RADIATION RETINOPATHY AFTER IODINE-125 BRACHYTHERAPY FOR UVEAL MELANOMA

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Purpose: Radiation retinopathy remains incompletely characterized and may cause severe vision loss. Ultra-wide-field fluorescein angiography provides a pan-fundus view of vascular alterations caused by radiation treatment and may predict visual and ocular outcomes. We have developed a grading scheme to describe pan-fundus severity and to predict the progression of radiation retinopathy in patients treated for uveal melanoma with iodine-125 brachytherapy.

Methods: A retrospective review of patients treated with standard iodine-125 brachytherapy for uveal melanoma at the Ophthalmic Oncology Center at the University of California, Los Angeles, who had undergone both baseline and postbrachytherapy ultra-wide-field fluorescein angiography. A grading scheme was devised based on observations of vascular leakage, retinal perfusion status, and retinal proliferation. The correlation of grade severity with patient characteristics, tumor features, visual acuity, optical coherence tomography findings, and neovascular glaucoma was measured with chi-square and one-way analysis of variance analyses.

Results: Sixty-seven patients were identified for review. Consistent wide-field angiographic patterns after brachytherapy were observed and graded as follows: Grade 0: normal; Grade 1: late foveal leakage; Grade 2: late peripheral leakage; Grade 3: presence of nonperfusion; and Grade 4: retinal neovascularization. Six eyes (8.9%) were Grade 0; 16 (23.8%) were Grade 1; 25 (37.3%) were Grade 2; 16 (23.4%) were Grade 3; and 4 (6.0%) were Grade 4. Higher grade radiation severity correlated significantly with duration of follow-up ($P < 0.02$); younger age ($P = 0.035$); worse visual acuity ($P = 0.001$); cystoid macular edema or atrophy on optical coherence tomography ($P < 0.0001$); and neovascular glaucoma ($P = 0.003$).

Conclusion: Wide-field fluorescein angiography revealed distinct fundus-wide patterns of vascular damage, which were progressive in nature in eyes treated with iodine-125 brachytherapy for uveal melanoma and correlated with signs of progressive vascular injury. This grading scheme may have prognostic value to predict the progression of radiation retinopathy and to prognosticate visual outcomes in patients undergoing brachytherapy.

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Although iodine-125 brachytherapy is the most common way to treat uveal melanoma in North America, vision-altering side effects from radiation are an established consequence.^{1,2} Radiation retinopathy, often with radiation-induced macular edema, remains the most challenging side effect with no long-term effective therapy.²

A reliable method for the accurate assessment of radiation retinopathy and its progression has yet to be fully described. Fluorescein angiography, optical coher-

ence tomography (OCT), and OCT angiography-based descriptions of radiation maculopathy have been reported.^{3–6} However, there have been no published reports describing the effects of radiation retinopathy beyond the macula.

We sought to assess the utility of ultra-wide-field fluorescein angiography in the evaluation and progression of radiation retinopathy after iodine-125 brachytherapy for local treatment of uveal melanoma. We evaluated ultra-wide-field fluorescein angiography to

seek consistent patterns of retinal vascular compromise. The findings were then categorized into a grading scheme reflecting progressive severity of the vasculopathy.

Methods

A retrospective case review was conducted in accordance with the tenets set forth in the Declaration of Helsinki and was approved by the Medical Institutional Review Board of the University of California, Los Angeles (UCLA). Patients diagnosed and treated for uveal melanoma with iodine-125 brachytherapy at the Ophthalmic Oncology Center at the Stein Eye Institute, UCLA, were reviewed. Uveal melanoma was diagnosed by clinical examination, ultrasonography, OCT, and fluorescein angiography. Standard iodine-125 plaque brachytherapy was performed according to the method described in previous reports from our institute.^{7–9}

Iodine-125 plaque brachytherapy was performed on all patients by T.A.M. The plaques were engineered by our institution's physics faculty within the Department of Radiation Oncology. The plaque shells consisted of a molded 1/100th inch thick gold foil with integrated anterior suture eyelets. Hemicircular notches were cut out of the shells on the opposite end from the suture eyelets. Iodine-125 seeds were affixed with cyanoacrylate in predetermined patterns specific to each plaque size. They were then covered by a layer of polymethyl methacrylate, with the interior surface conformed to a globe of radius 25 mm. Dose was calculated as per the Collaborative Ocular Melanoma Study protocol using the American Association of Physicists in Medicine TG-43U1 methodology.^{10,11} Because the plaques used nearly water equivalent–molded polymethyl methacrylate instead of the silicone elastomer insert used by Collaborative Ocular Melanoma Study plaques, the plaques were not subjected to the dose reduction of approximately 10% because of the silicone elastomer transmission factor that has been reported for Collaborative Ocular Melanoma Study plaques.⁵

Dose rates were within the acceptable dose rate range reported in the Collaborative Ocular Melanoma Study approximately 50 cGy/hour. Plaques were built

to encompass the widest basal dimension of the tumor with 3 mm lateral margins. The prescription dose was 85 Gy to the apical height of the tumor.

In brief, the four rectus muscles were identified and looped with silk sutures; the location of the tumor on the sclera was identified and marked with transpupillary transillumination; transscleral fine needle aspiration biopsy was performed for cytopathology and molecular prognostication; and the plaque was sutured and its position confirmed with intraoperative ultrasonography.¹² After treatment, patients were evaluated with clinical examination, ultrasonography, color photography, wide-field fluorescein angiography, and OCT in a standard follow-up protocol every 6 months, except when severe media opacity precluded a view of the fundus. Patient demographics, tumor characteristics, OCT, and Optos ultra-wide-field fluorescein angiography were collected and reviewed during the month of June 2015.

To be included in this investigation, patients were required to have had a baseline ultra-wide-field fluorescein angiogram before brachytherapy and at last follow-up (Optos, Inc, Dunfermline, United Kingdom). Wide-field fluorescein angiography became available at our institution in 2005; however, its use was limited to mostly peripheral tumors. Furthermore, routine 6 monthly wide-field angiography after brachytherapy was not performed until 2012. Patients were excluded from the review for the following reasons: preexisting history of glaucoma; history of diabetes mellitus; media opacity (e.g., cornea opacity, cataract, and vitreous hemorrhage) preventing adequate image quality of the posterior segment; and eyes with baseline macular or peripheral angiographic abnormality.

The fluorescein angiographic findings of the entire fundus were evaluated and recorded. Six minutes or later after intravenous dye injection was defined as late phase. Fluorescein angiographic pattern of the actual treated choroidal or ciliary body melanoma itself was not recorded. A grading scheme was created based on the observations. Two independent examiners (T.A.M. and E.K.) who were masked to the clinical outcomes reviewed deidentified patient images and assigned a grade based on angiographic findings.

Grades were correlated with the time from brachytherapy to the date of the wide-field fluorescein; we made the assumption that the grade assigned occurred on the date the angiogram was obtained. Grades were correlated with certain patient demographics, tumor characteristics, vision outcomes, and progressive radiation maculopathy identified with OCT. Five macular tumors were excluded from these analyses to minimize the confounding effect of tumor influence on vision and macular OCT findings, which may be distinct from radiation vasculopathy.

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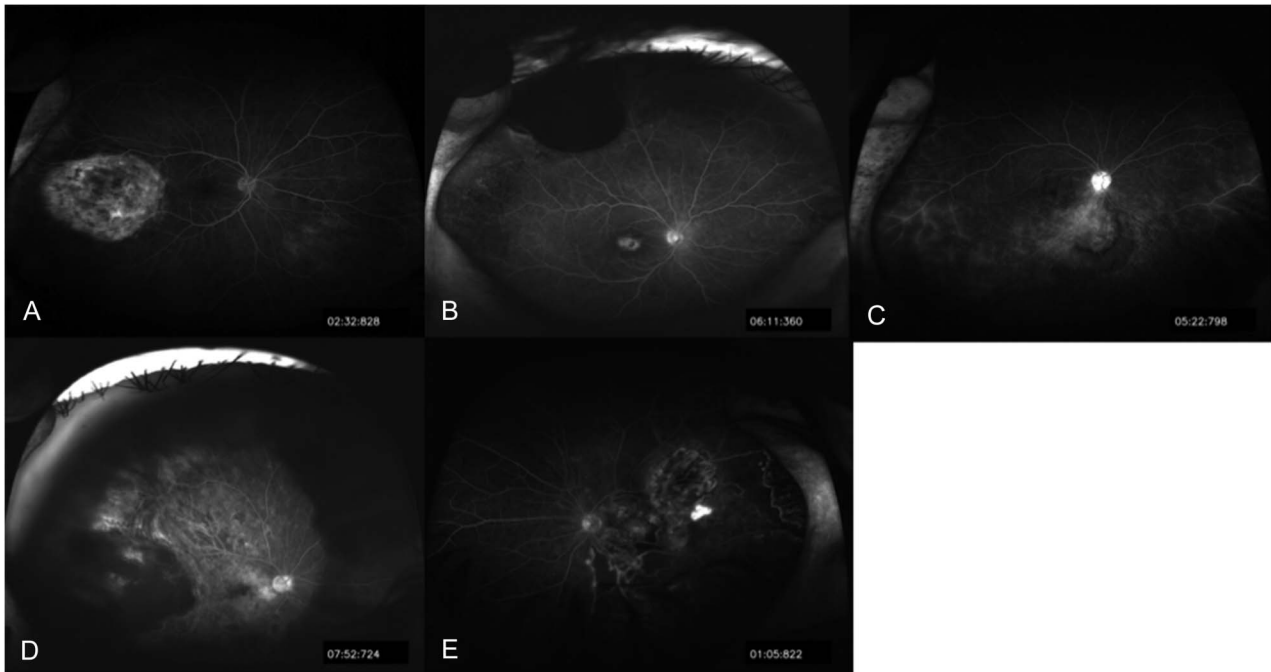


Fig. 1. Ultra-wide-angle Optos fluorescein angiography in patients treated with iodine-125 brachytherapy demonstrating examples of the grades of radiation severity. **A.** Grade 0, no vascular abnormality outside the tumor (mild nonspecific choroidal stain inferonasally, not retinal vascular leakage). **B.** Grade 1, foveal leakage. **C.** Grade 2, peripheral leakage. **D.** Grade 3, nonperfusion. **E.** Grade 4, retinal neovascularization.

Although radiation papillopathy may also impact visual outcomes, it is difficult to quantitate and was not included in this grading system.

All statistical analyses were conducted with IBM SPSS, version 22.0 (Chicago, IL). Pairwise comparisons with one-way analysis of variance were used for the comparison of time from brachytherapy treatment in Grades 1 to 4. The chi-square test was used to analyze the difference between nominal parameters, and one-way analysis of variance was used for comparisons of continuous parameters among different grades.

Results

Sixty-seven patients met the study criteria for inclusion. There were 56 choroidal, 6 ciliary body, and 5 iris melanoma cases. Consistent fluorescein

angiographic findings of the fundus were observed to varying degrees: foveal leakage, peripheral vascular leakage, nonperfusion, and retinal neovascularization. In cases with more extensive radiation vasculopathy, these findings were found in combination. A grading scheme was devised and a grade assigned to each image. A two-person independent validation revealed agreement in angiographic findings. A discrepancy in grading was revealed in one case; both individuals reviewed the imaging together and determined the grading severity in a consensual manner.

Ultra-Wide-Field Fluorescein Angiographic Grading Scheme

Grade 0. Patients with Grade 0 demonstrated normal fluorescein angiographic findings except for the melanoma (Figure 1, A–E).

Table 1. Number of Eyes for Each Grade of Radiation Retinopathy, Mean ($\pm$ SD) Postoperative Years From Iodine-125 Plaque Implantation to the Development of Radiation Retinopathy and *P* Values From the Pairwise Comparison Using One-Way Analysis of Variance

	No. of Eyes	Postoperative Years	<i>P</i> From Pairwise Comparison			
			Grade 1	Grade 2	Grade 3	Grade 4
Grade 1	15	1.6 $\pm$ 1.1	NA	0.006	0.000	0.000
Grade 2	29	3.1 $\pm$ 1.4	0.006	NA	0.027	0.002
Grade 3	18	4.2 $\pm$ 1.5	0.000	0.027	NA	0.159
Grade 4	4	5.8 $\pm$ 0.9	0.000	0.002	0.159	NA

NA, not applicable.

Grade 1. Patients with Grade 1 radiation vasculopathy demonstrated any degree of foveal leakage in the late frame of fluorescein angiography.

Grade 2. Patients with Grade 2 radiation vasculopathy demonstrated peripheral vascular leakage in addition to foveal leakage in the late frame of fluorescein angiography.

Grade 3. Patients with Grade 3 radiation vasculopathy demonstrated nonperfusion, defined by a greater than one disk area of lack of retinal vascular filling in

the midphase in addition to late foveal and peripheral leakages.

Grade 4. Patients with Grade 4 radiation vasculopathy demonstrated retinal neovascularization; these patients also had concomitant midphase nonperfusion and late foveal and peripheral leakages.

Grading Severity and Clinical Associations

Table 1 demonstrates the correlation of retinopathy grade with time since iodine-125 brachytherapy. Table 2

Table 2. Grading Scheme Severity Correlations With Patient Characteristics, Tumor and Related Treatment Facts, and Vision-Related and Ischemic Outcomes

	Grade 0 (n = 6)	Grade 1 (n = 16)	Grade 2 (n = 25)	Grade 3 (n = 16)	Grade 4 (n = 4)	Total (n = 67)	P
Patient characteristics							
Age, years (mean ± SD)	59.6 ± 7.3	71.5 ± 13.1	60.2 ± 13.9	59.5 ± 16.0	51.6 ± 12.8	62.1 ± 14.5	0.035*
Sex (male:female)	4:2	7:9	13:12	7:9	1:3	32:35	0.722
Ethnicity (whites: nonwhites)	3:3	12:4	20:5	9:7	1:3	45:22	0.126
Laterality (right:left)	2:4	7:9	13:12	9:7	0:4	31:36	0.302
Hypertension (no:yes)	2:4	12:4	13:12	14:2	2:2	43:24	0.060
Smoking (packyears, mean ± SD)	10.4 ± 15.0	12.1 ± 20.6	4.6 ± 9.3	5.4 ± 12.1	10.0 ± 11.5	7.4 ± 14.0	0.487
Lens status (phakic: pseudophakic)	6:0	10:6	22:3	15:1	4:0	57:10	0.054
Tumor- and treatment-related facts							
Muscle hangback during surgery (no:yes)	2:4	10:6	15:10	12:4	4:0	43:24	0.219
Presence of subretinal fluid from preoperative ultrasonography (no: yes)	0:3	7:9	11:14	3:11	2:2	23:39	0.362
Tumor location (choroidal:iris:ciliary body)	5:1:0	12:2:2	20:2:3	15:0:1	4:0:0	56:5:6	0.785
Involvement of ciliary body (no:yes)	4:2	11:5	19:6	14:2	4:0	52:15	0.520
Tumor height, mm (mean ± SD)	5.1 ± 3.2	5.6 ± 2.6	4.8 ± 2.4	5.6 ± 2.7	4.3 ± 3.4	5.11 ± 2.6	0.875
Size of the plaques, mm (mean ± SD)	15.5 ± 1.7	18.3 ± 4.2	17.3 ± 2.7	17.0 ± 2.8	16.4 ± 1.8	17.3 ± 3.1	0.359
Vision-related and ischemic outcomes							
Best-corrected visual acuity, logMAR (mean ± SD) [Snellen equivalent]	(0.2 ± 0.1) [20/32]	(0.6 ± 0.6) [20/80]	(0.9 ± 0.9) [20/159]	(1.4 ± 0.8) [20/502]	(1.8 ± 0.5) [20/1262]	(0.9 ± 0.9) [20/159]	0.001*
ETDRS letter change from baseline†	5	10	30	40	80	25	0.021‡
Macular morphology with OCT (normal contour:CME:CME + atrophy:atrophy)	6:0:0:0	8:8:0:0	10:11:4:0	2:6:3:5	0:0:3:1	26:25:10:6	<0.0001*
Neovascular glaucoma (no:yes)	6:0	16:0	25:0	16:0	3:1	66:1	0.003*

Chi-Square test was used for nominal variables, and one-way analysis of variance test was used for continuous variables.

*Statistically significant.

†Early Treatment Diabetic Retinopathy Study letter change based on logMAR conversion of Snellen equivalent.

‡LogMAR equivalent (not shown) used to determine statistical significance.

CME, cystoid macular edema; logMAR, logarithm of the minimum angle of resolution; ETDRS, Early Treatment Diabetic Retinopathy Study.

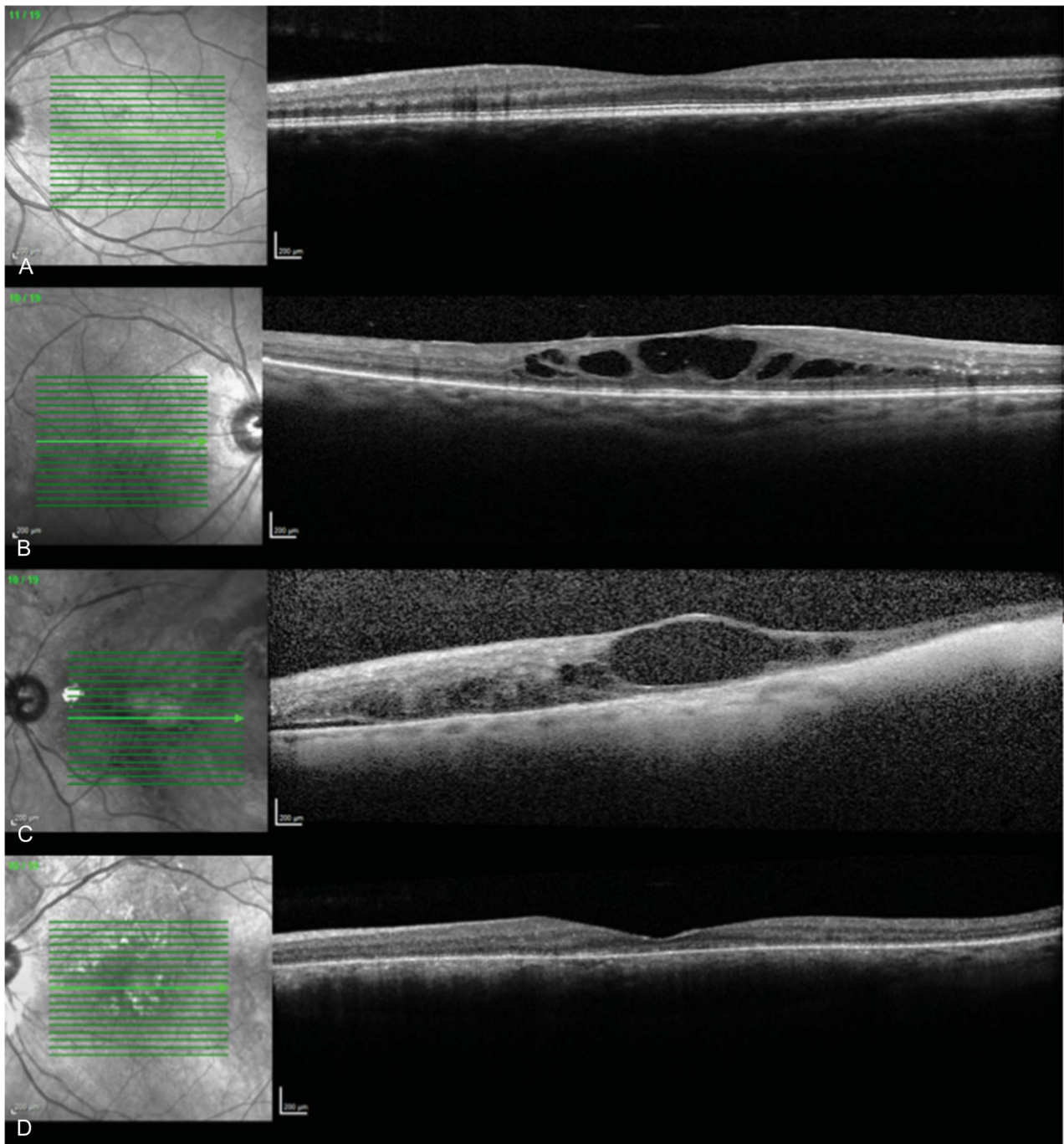


Fig. 2. Optical coherence tomography of the macula in patients treated with iodine-125 brachytherapy demonstrating examples of the natural history of progressive radiation maculopathy. **A.** Normal foveal contour. **B.** Cystoid macular edema. **C.** Cystoid macular edema and atrophy. **D.** Atrophy.

demonstrates the correlation of grade with the following categories: patient factors, tumor-related factors, visual outcomes, OCT macular findings, and neovascular glaucoma. Figure 2, A–D demonstrates the course of macular OCT progression in radiation retinopathy.

Discussion

We describe the characteristics of ultra-wide-field fluorescein angiographic radiation vasculopathy in eyes with uveal melanoma that was treated with iodine-125 brachytherapy. We also established an

ultra-wide-field fluorescein angiographic grading scheme, whose severity correlates with time from treatment, younger patient age, visual decline, macular OCT progression in radiation retinopathy, and neovascular glaucoma.

Ultra-wide-field fluorescein angiography has enhanced our understanding of, and may improve our management of, vascular diseases such as diabetic retinopathy, vein occlusions, and uveitis.^{13–16} Unlike standard fundus photography and fluorescein angiography, ultra-wide-field fluorescein angiography in eyes with treated uveal melanoma provides simultaneous evaluation of the macula and the periphery as well as the tumor itself. Vascular evaluation of the entire fundus may allow for predicting the progression of radiation vasculopathy and may alter how we manage peripheral abnormalities.

The grading scheme severity correlated with features of progressive ischemia. We report that the grading severity correlated with visual decline, progressive maculopathy on OCT, and development of neovascular glaucoma. This underscores the fact that radiation retinopathy is an ischemic vascular disease with a predictable pattern of progression that may involve the entire fundus.

There was an association of retinopathy grade and younger patient age. The trend toward phakic lens status and more severe retinopathy grades may reflect younger patient age. Gragoudas et al¹⁷ reported an association with young age and development of proliferative radiation retinopathy after proton beam therapy of uveal melanoma. Bianciotto et al¹⁸ also reported younger age as a risk factor for the development of proliferative radiation retinopathy after iodine-125 brachytherapy. Bianciotto's study reported a 5.8% rate of neovascularization in 5 years. This relatively small proportion of patients with retinal neovascularization may explain why we did not see a statistically significant difference in time from treatment between Grade 3 and Grade 4 cases (Table 1); perhaps, it takes long time to progress from Grade 3 to 4. Although the reason for increased susceptibility to proliferative radiation retinopathy in younger aged patients is not well understood, this may be related to a differential vascular endothelial growth factor response in younger aged patients in the setting of hypoxia. It has been shown that reduced hypoxia-inducible factor-1 activity in old-aged rabbits compared with young-aged rabbits could explain age-related impairment of angiogenesis in response to ischemia.¹⁹ In addition, we made the assumption that the time of angiogram acquisition was the date of onset of the grade classification, when in fact the patient could have acquired the findings within 6 months before the angiogram was obtained. Despite

this limitation imposed by the time intervals between clinical follow-up, we demonstrated a statistically significant difference in time from treatment for most of the pairwise comparisons between grades.

The limitations of this study are the retrospective nature of the data collection, the unequal time intervals between visits, the lack of consistent protocols for imaging acquisition, and the small number of cases, which could affect statistical power. For example, there were only 4 patients who were categorized as Grade 4 radiation retinopathy and only one patient who developed neovascular glaucoma. Furthermore, we did not include specific data on individual patient progression, but rather compared snapshots in time for different patients, each who may have varying parameters that influence radiation retinopathy progression. Finally, the impact of brachytherapy on the choroid may likely contribute to the clinical findings and visual outcomes in patients with radiation vasculopathy; however, our study did not include indocyanine green angiography, which might better characterize this.

In summary, we have demonstrated that radiation vasculopathy after iodine-125 brachytherapy affects the entire fundus. We describe a new grading scheme, which can be used to identify and stage radiation retinopathy, and may help identify early ischemic complications from radiation treatment. Finally, this grading scheme can be used as a discrete outcome in clinical research to better understand and treat radiation vasculopathy.

Key words: uveal melanoma, choroidal melanoma, ciliary body melanoma, brachytherapy, iodine-125 plaque, radiation retinopathy, fluorescein angiography, Optos wide-field imaging, grading scheme.

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